LITERATURE REVIEW

Soft robotics has emerged as an important research area for developing robotic systems that can safely interact with humans and objects with different shapes, sizes, weights, and levels of fragility. Unlike conventional robotic mechanisms that primarily rely on rigid links and joints, soft robotic systems make use of compliant materials and flexible structures to achieve deformation and motion. This approach is particularly useful in applications such as wearable rehabilitation devices, prosthetic systems, and robotic grippers, where mechanical compatibility, safe interaction, and adaptability are important requirements.

The literature reviewed for the present work mainly focuses on soft actuation techniques, compliant robotic grippers, smart materials, and the integration of sensing capabilities into soft structures. The studies also highlight the trade-offs between force generation, compliance, controllability, portability, and sensing. These aspects are particularly relevant to the development of a robotic grasping system, where the actuator must generate appropriate motion while the sensing system must provide information about deformation, force, or interaction with the grasped object.

2.1 Soft Robotics and Alternative Actuation Techniques

The author conducted a review titled *A Review of Soft Robotics and Soft Rehabilitation Gloves: Exploring Alternative Soft Robots Actuation Techniques*. The work investigates the development of soft robotic systems, with particular emphasis on wearable assistive devices and rehabilitation gloves. The study highlights the limitations of conventional rigid robotic mechanisms when they are required to interact closely with the human body. Traditional mechanisms generally use materials with high Young's modulus and rigid joints, which can create difficulties in achieving compatibility with the natural motion of human joints.

Soft robotics addresses this issue by replacing conventional rigid components with compliant elastomeric structures. The use of soft actuators allows the structure itself to deform during operation, reducing the requirement for precisely aligning the robot's centre of rotation with the biological joint. This is particularly beneficial for rehabilitation applications, where the motion of the robotic device must remain compatible with the user's natural movement.

The study discusses several alternative actuation technologies. Silicone elastomers such as Ecoflex and Dragon Skin are used in pneumatic structures in which internal chambers expand anisotropically when pressurized. The deformation of these chambers can then be used to produce bending or other desired movements. In addition to pneumatic actuation, smart materials such as Shape Memory Alloys (SMAs) and Dielectric Elastomer Actuators (DEAs) are considered as artificial-muscle-type actuators that respond to thermal or electrical stimulation.

One of the important observations from the study is that soft robotic systems provide significant advantages in terms of compliance and adaptability, but their deformation characteristics also introduce challenges in control. Soft materials can exhibit considerable hysteresis and nonlinear

deformation, making conventional control methods such as traditional PID control more difficult to implement accurately. The performance of the actuator is therefore closely related to the nonlinear behaviour of the material.

The review also identifies portability and durability as important challenges. Pneumatic systems generally require external compressors, valves, and associated hardware, which can increase the overall size of the system and reduce its portability. Repeated actuation can also result in fatigue and material degradation, affecting the long-term reliability of soft actuators.

For future development, the study points toward more compact and self-contained actuation systems, including electro-hydraulic approaches. It also emphasizes the integration of flexible and stretchable sensors, such as liquid-metal strain gauges, directly into elastomeric structures. Such integration can provide real-time information about the deformation of the soft actuator and can potentially support proprioception and force-feedback capabilities.

The study also discusses tendon-driven actuation as an alternative approach for soft robotic systems. In tendon-driven mechanisms, electric motors pull cables routed through soft sheaths or guides along the fingers. The resulting cable tension forces the soft finger structure to curl inward. One major advantage of this arrangement is that the relatively heavy motors can be positioned away from the gripping region. This reduces the distal weight of the gripper while still allowing relatively precise position control.

However, tendon-driven systems introduce their own limitations. Friction between the cable and its routing mechanism can affect force transmission, while the requirement for appropriate cable routing increases mechanical complexity. Compared with fluidic actuation, tendon-driven mechanisms may also provide less intrinsic compliance.

Shape Memory Alloys are another important technology discussed in the review. SMA wires, such as Nitinol, contract when heated, usually through electrical current, and can therefore produce a pulling action similar to biological muscle contraction. Their high power-to-weight ratio, compact form, silent operation, and simple structural arrangement make them attractive for soft robotic applications. However, the relatively slow cooling process limits their response speed. They can also require considerable energy for heating and exhibit nonlinear hysteresis.

Overall, this study establishes that there is no single actuation technology that provides an ideal combination of force, response speed, portability, compliance, and controllability. The selection of the actuator must therefore depend on the intended robotic application and its operating requirements.

2.2 Robotic Gripper Design and Compliant Mechanisms

The author presented a systematic review titled *Current Designs of Robotic Arm Grippers: A Comprehensive Systematic Review*. The study evaluates different robotic arm gripper configurations and investigates their suitability for handling objects with varying fragility,

geometry, and weight. The review categorizes existing grippers according to mechanical mobility, degrees of freedom (DOF), and actuation mechanism.

Mechanical mobility is classified into completely constrained, underconstrained, and deformable configurations. This classification provides a way to understand how the mechanical structure of a gripper determines its ability to adapt to objects. The number of degrees of freedom is also considered because it determines the number of independent motions available to the gripper. In addition, the actuation mechanism determines how these motions are generated and controlled.

The review highlights a transition from conventional rigid two- and three-jaw grippers toward compliant and soft robotic mechanisms. Conventional rigid grippers are capable of generating relatively high forces and are suitable for heavy objects, but they can be less effective when handling fragile or irregularly shaped objects. Soft and compliant structures, on the other hand, can deform around an object and provide a more adaptive grasp.

Several material and fabrication approaches are discussed, including Thermoplastic Elastomers (TPEs), Dielectric Elastomer Actuators (DEAs), and flexible polymers manufactured using 3D printing. Monolithic compliant mechanisms are particularly important because they generate motion through elastic deformation rather than through multiple rigid components and conventional mechanical hinges. This can reduce sliding friction and mechanical backlash within the mechanism.

The use of 3D printing and topological optimization also provides opportunities to design structures with controlled flexibility and complex geometries. High-elasticity materials allow the gripper to passively adapt to different object shapes without requiring a separate control action for every deformation. This concept is described through the idea of "embodied intelligence," where part of the intelligence of the system is incorporated into its physical structure rather than being implemented entirely through software and control algorithms.

An important aspect of this study for the present work is the integration of sensing into compliant mechanisms. The review discusses the direct integration of piezoresistive strain gauges into flexible structures. One reported configuration achieved a force of 16 mN within a 2.2 mm workspace. The integration of sensing elements directly into the compliant structure provides a method for obtaining information about mechanical deformation and applied force.

This approach is particularly relevant to soft robotic grasping because the sensing system can provide information about the state of the deformable structure without requiring a large number of conventional rigid sensors. At the same time, the study identifies an important limitation: integrating sensors into a compliant mechanism can impose restrictions on the operating workspace and overall mechanical behaviour.

The review also identifies limitations associated with the mechanical performance of compliant structures. Since compliant links are inherently flexible, they generally have lower torque

capability than rigid metallic links. This limits the payload that can be handled by the gripper. Although soft mechanisms provide good adaptability for fragile objects, they may also have difficulty performing complex in-hand manipulation tasks when compared with highly sensorized, multi-fingered rigid robotic hands.

The literature therefore indicates a fundamental trade-off between compliance and force capability. A highly compliant structure can adapt naturally to fragile and irregular objects, whereas a rigid structure can provide greater force and payload capability. Current research consequently focuses on developing mechanisms that can retain the adaptability of soft structures while improving their force and torque generation capabilities.

2.3 Sensing in Soft and Compliant Robotic Systems

The reviewed literature indicates that sensing is an important component in achieving effective control of soft robotic systems. Unlike rigid mechanisms, soft structures undergo continuous deformation, making it difficult to determine their configuration using only conventional position measurements. Information regarding deformation, force, and interaction with an object can therefore be important for controlling the grasping process.

The author highlight the potential of flexible and stretchable sensing technologies integrated directly into elastomeric actuator structures. Liquid-metal strain gauges are identified as one possible approach for obtaining real-time proprioceptive information and force feedback. Since the sensor can be incorporated into the soft substrate, the sensing mechanism can deform along with the actuator instead of acting as a separate rigid component.

Similarly, The author discuss the integration of piezoresistive strain gauges within compliant robotic structures. Such sensors can respond to deformation of the flexible mechanism and provide information related to applied force. The reported example of a 16 mN force within a 2.2 mm workspace demonstrates the possibility of combining sensing and compliant mechanical structures.

These approaches are significant because sensing in a soft gripper cannot be considered independently of the mechanical design. The sensor placement, deformation characteristics of the substrate, and available workspace can all influence the quality and usefulness of the measurement. Direct sensor integration can simplify the overall mechanical arrangement, but it may also introduce constraints on the deformation range.

The literature therefore suggests that an effective soft robotic sensing system should be considered as an integrated part of the actuator and gripper design rather than as an independent component added after the mechanical system has been developed.

2.4 Comparison and Synthesis of the Reviewed Studies

The two reviewed studies approach soft robotic systems from complementary perspectives. The author primarily focus on soft robotics and alternative actuation technologies, particularly in the context of wearable and rehabilitation systems. The author focus more broadly on robotic gripper architectures and the mechanical principles that determine their ability to handle different objects.

A common observation between the studies is the importance of compliance. Soft elastomers, flexible polymers, compliant mechanisms, SMAs, DEAs, pneumatic structures, and tendon-driven systems provide different methods for producing motion while allowing some degree of adaptability. This adaptability is especially useful when the robotic system must interact with fragile or irregularly shaped objects.

The studies also demonstrate that the selection of an actuation method involves several trade-offs. Pneumatic actuation can provide compliant motion but generally requires external compressors and valves. Tendon-driven systems can provide precise movement while keeping the motors away from the gripping region, but cable friction and routing complexity can affect performance. SMAs provide a compact and lightweight actuation solution with a high power-to-weight ratio, but their response is limited by thermal behaviour and hysteresis. DEAs and other smart-material approaches provide another route toward artificial-muscle-like actuation, although their implementation introduces its own design considerations.

From the sensing perspective, both studies indicate a movement toward integrating sensing elements into flexible structures. This is important because the deformation of a soft actuator contains information about its state and interaction with the environment. Rather than depending only on external rigid sensors, flexible sensing elements can potentially provide information directly from the deforming structure.

The studies also show that improving adaptability often comes at the expense of force, workspace, or control simplicity. Compliant grippers are effective at safely handling fragile objects but may have lower torque and payload capabilities. Similarly, the nonlinear and hysteretic behaviour of soft materials makes accurate control more challenging.

2.5 Limitations and Identified Research Gaps

The reviewed literature identifies several unresolved challenges in the development of soft robotic systems.

First, the nonlinear deformation and hysteresis associated with soft materials make accurate modelling and control difficult. Conventional control approaches may not be sufficient to describe the complete behaviour of highly deformable actuators. This creates a requirement for suitable sensing and control strategies that can account for the actual deformation of the system.

Second, portability remains a significant issue for pneumatic systems. The requirement for external compressors and valves can make the complete system bulky, particularly for wearable or compact robotic applications. This motivates research into self-contained actuation systems.

Third, long-term durability is an important concern. Repeated deformation can result in fatigue and degradation of elastomeric materials, potentially affecting both actuator performance and integrated sensing elements.

From the gripper perspective, compliant mechanisms face limitations in output torque and payload capacity. Although their ability to adapt to fragile objects is advantageous, they may not provide the force required for heavier objects or complex manipulation tasks. There is therefore a need for designs that combine compliance with improved force and torque capability.

The literature also shows that integrated sensing can introduce mechanical limitations. For example, the integration of piezoresistive sensing into a compliant mechanism may restrict the available workspace. This indicates that sensor selection and placement must be considered together with the mechanical and actuation design.

A broader research direction emerging from the reviewed studies is the development of robotic systems in which actuation, mechanical compliance, and sensing are designed together. Rather than treating the actuator, gripper structure, and sensor as completely independent components, an integrated design can potentially improve adaptability, feedback, and overall system performance.

2.6 Relevance to the Present Project

The reviewed literature provides a direct foundation for the Sensors aspect of the present project. The studies demonstrate that sensing is particularly important in soft and compliant robotic systems because the mechanical state of a deformable structure cannot always be determined through conventional rigid-body position measurements.

The use of flexible strain-based sensing and the integration of sensing elements into compliant structures provide relevant approaches for detecting deformation and estimating interaction forces. These techniques can support real-time feedback during grasping and can help the robotic system respond according to the state of the soft structure and the object being handled.

At the same time, the literature emphasizes that sensor integration cannot be separated from the actuator and gripper design. Sensor placement, deformation range, workspace, sensitivity, and the mechanical behaviour of the substrate need to be considered together. The limitations reported in the reviewed studies therefore provide useful design considerations for the present project.

The literature also supports the need for a balance between compliance, sensing capability, force generation, and controllability. A suitable sensing system should provide meaningful

information about the gripper's deformation or interaction while maintaining the required mechanical range of motion.

Therefore, the reviewed work establishes the technical motivation for investigating sensing approaches suitable for a soft robotic grasping system. The present project can build on these findings by considering how sensing can be incorporated into the robotic mechanism to obtain useful feedback during actuation and grasping, while accounting for the limitations associated with flexible materials, nonlinear deformation, sensor placement, and restricted workspace.

2.7 Summary of Literature Review

The literature reviewed demonstrates a clear transition from conventional rigid robotic mechanisms toward soft, compliant, and smart-material-based systems. The author examined alternative soft-robotic actuation techniques including pneumatic/fluidic actuation, tendon-driven mechanisms, Shape Memory Alloys, and Dielectric Elastomer Actuators. The study highlights the advantages of these technologies in terms of compliance, compactness, lightweight construction, and compatibility with human motion, while also identifying challenges related to nonlinear behaviour, hysteresis, portability, energy consumption, response time, and durability.

The author reviewed robotic arm grippers and emphasized the role of mechanical mobility, degrees of freedom, actuation mechanisms, and compliant structures in handling objects with different shapes and fragility. The work highlights the potential of soft and 3D-printed compliant mechanisms to achieve passive adaptation and embodied intelligence. It also discusses integrated piezoresistive sensing and demonstrates how sensing can be incorporated into flexible structures.

A common research direction across the literature is the integration of actuation, compliance, and sensing into a unified robotic structure. However, significant challenges remain in achieving high force capability, large workspace, accurate sensing, portability, durability, and reliable control simultaneously.

For the present project, the most relevant outcome of the literature survey is that sensing must be considered as an integral part of the soft robotic mechanism. The reviewed approaches provide a basis for investigating suitable sensing methods for detecting deformation and grasp-related interaction, while the identified limitations provide important considerations for the design and implementation of the proposed system.